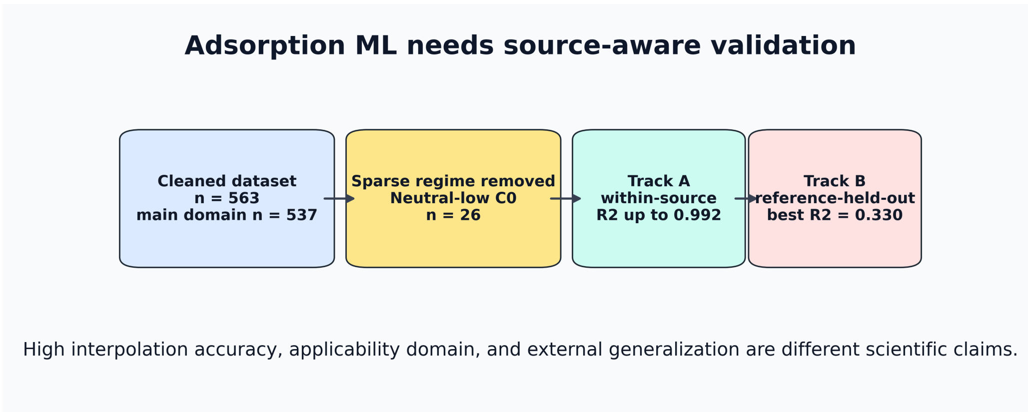
Graphical Abstract



Highlights

- Source-specific adsorption models reached R^2 up to 0.992.
- Sparse neutral-low-concentration cases were excluded from primary modeling.
- Source-origin heterogeneity explained validation sensitivity.
- Row-wise validation overstated source-transfer performance.
- Reference-held-out sensitivity reached $R^2 = 0.330$ in the main domain.

Source-aware machine learning for AMD-relevant metal adsorption: distinguishing high-accuracy interpolation from cross-source generalization

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Abstract

Acid mine drainage (AMD) is a persistent water-quality challenge because acidic, sulfate-rich drainage mobilizes metal and metalloid contaminants. Adsorption is widely investigated for AMD-relevant contaminant remediation, and machine learning (ML) can support rapid prediction of removal efficiency; however, reported accuracy can depend strongly on feature choice, applicability-domain coverage, and validation design. This study develops a source-aware and leakage-aware validation framework for adsorption-based metal-contaminant removal-efficiency prediction using a literature-compiled experimental dataset. Source awareness was introduced by tracking the literature reference from which each adsorption record originated. Leakage awareness was addressed by testing whether model performance remained reliable when familiar rows, adsorbents, or entire literature sources were removed from training, and by separating capacity-inclusive process-monitoring models from capacity-free pre-experiment screening

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models. The raw dataset contained 566 adsorption records; after deterministic numeric parsing and quality filtering, 563 valid records remained. Because the neutral/low-concentration regime contained only 26 records, primary modeling was restricted to a 537-record domain covering acidic/high-concentration, acidic/low-concentration, and neutral/high-concentration conditions. Track A quantified source-specific interpolation through within-source cross-validation, where XGBoost reached R^2 values of 0.992, 0.974, and 0.960 for three major source groups. Track B tested broader generalization using random row-wise, adsorbent-held-out, and reference-held-out splits. In the restricted primary domain, the full capacity-inclusive LightGBM model achieved R^2 = 0.806 under random row-wise validation. Under stricter validation, the best adsorbent-held-out performance was obtained by a capacity-free numeric LightGBM model (R^2 = 0.532), while reference-held-out performance reached R^2 = 0.330 for a capacity-inclusive LightGBM model and R^2 = 0.305 for the best capacity-free model. A source-domain audit showed that source-origin heterogeneity strongly affected validation sensitivity. These results support ML-assisted prediction for adsorption-based AMD-relevant metal removal, but broader deployment requires explicit applicability-domain definition, careful separation of feature-use cases, and validation against novel adsorbents and literature sources.

Keywords: acid mine drainage, adsorption, metal removal efficiency, machine learning, XGBoost, LightGBM, source-aware validation

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1. Introduction

Mining activities are a major contributor to water-quality degradation, particularly through the generation of acid mine drainage (AMD). AMD is produced mainly through oxidative dissolution of sulfide minerals and is commonly associated with low pH, high sulfate concentration, and elevated dissolved metal or metalloid concentrations [1]. These conditions can mobilize contaminants such as Fe, Mn, Pb, Cu, Zn, Cr, As, Cd, and Ni, creating long-term risks for receiving water bodies, soils, aquatic ecosystems, and downstream water reuse. Because AMD chemistry varies across mine sites, ore bodies, weathering history, and treatment conditions, robust prediction of contaminant removal remains a difficult environmental-chemical-engineering problem.

Adsorption is widely investigated for AMD-relevant contaminant remediation because it can operate under mild conditions, can use low-cost or waste-derived adsorbents, and can be optimized through pH, adsorbent dose, contact time, temperature, surface chemistry, and initial contaminant concentration. Recent reviews emphasize adsorption as an active research direction for AMD treatment, including carbonaceous, mineral, oxide, biomass-derived, and composite adsorbents [2]. The expansion of experimental adsorption datasets has therefore encouraged the use of machine learning (ML), a subset of artificial intelligence, to predict removal efficiency and accelerate adsorbent or operating-condition screening.

ML is attractive in this setting because adsorption performance depends on nonlinear interactions among solution chemistry, contact time, adsorbent dose, contaminant identity, material surface area, and source-specific synthesis or reporting practices. Data-driven models can capture multivariate trends that are difficult to express with a single empirical equation, and they can provide rapid screening support once trained on reliable data. However, the validity of an ML model depends not only on the algorithm but also on whether the test data are genuinely independent of the training data and whether the input features are available at the intended stage of deployment.

Recent adsorption and environmental-remediation ML studies demonstrate that high predictive accuracy is possible (Table 1). Hafsa et al. modeled heavy-metal adsorption efficiencies using multiple ML algorithms and reported R^2 values in the 0.96–0.99 range under ten-fold cross-validation, with experimental data supplemented by synthetic interpolation [3]. Lu et al. predicted heavy-metal removal by chitosan-based flocculants using random forests and reported $R^2 = 0.9354$, identifying solution pH and flocculant molecular weight as important drivers [4]. Wang et al. used interpretable ML for heavy-metal removal by biochar and reported XGBoost test performance near $R^2 = 0.99$, together with SHAP and partial-dependence analyses to relate removal efficiency to biochar and operating variables [5, 8]. Recent ensemble-learning work on biochar adsorption similarly reported strong XGBoost performance for heavy-metal adsorption efficiency prediction [6]. Within the scope of *Journal of Environmental Chemical Engineering*, interpretable XGBoost modeling has also been applied to heavy-metal removal efficiency in electrokinetic remediation, demonstrating the journal relevance of transparent ML workflows for environmental treatment systems [7].

The central challenge is that high validation accuracy does not necessarily imply reliable generalization to unseen laboratories, adsorbents, contaminant systems, or literature sources. In many adsorption ML studies, performance is estimated using random row-wise validation, where individual experimental records are randomly divided between training and testing. This can be useful for measuring interpolation within a compiled dataset, but it may also place closely related records from the same study dataset, adsorbent series, or operating campaign in both the training and test sets. As a result, the test set can appear independent while

Table 1: Representative adsorption/remediation ML studies and validation gaps motivating the present work.

Study/domain	Reported strength	Gap addressed here
Heavy-metal adsorption ML [3]	High cross-validation R^2 values, including synthetic interpolation support	Cross-validation can indicate interpolation performance but may not test transfer to unseen literature sources or adsorbent campaigns.
Chitosan-based flocculant removal [4]	Random-forest prediction with $R^2 > 0.93$ and interpretable pH/molecular-weight effects	Strong within-domain prediction does not by itself establish source-held-out or material-held-out generalization.
Biochar heavy-metal adsorption [5, 6]	XGBoost/ensemble models with high test performance and interpretable feature effects	Adsorbent families, feedstocks, and protocols may appear in both train and test sets under row-wise validation.
Electrokinetic remediation ML in JECE [7]	Demonstrates journal relevance of explainable ML for contaminant-removal efficiency	Broader adsorption workflows still need explicit feature-use-case separation and source-aware validation.
PFAS adsorption/partitioning ML [11]	Random-split performance substantially exceeded leave-compound-out transfer	Supports the need to distinguish interpolation from transfer to unseen chemical/source domains.

still containing familiar source-specific patterns. In ML-based science, such data leakage and validation mismatch can produce overoptimistic estimates of deployable performance [9]. Environmental ML best-practice guidance similarly emphasizes careful data splitting, leakage management, randomness assessment, and evaluation design [10]. These concerns are particularly important for adsorption datasets because different studies may use different adsorbent synthesis routes, contaminant species, concentration ranges, analytical protocols, and reporting conventions.

Feature definition can further blur the interpretation of reported accuracy. Adsorption capacity and removal efficiency are both derived from concentration changes in many batch adsorption experiments:

$$\mathrm{RE}(\%) = 100\frac{C_0 - C_e}{C_0}, \tag{1}$$

$$q_e = \frac{(C_0 - C_e)V}{m}, \tag{2}$$

where C_0 and C_e are the initial and equilibrium concentrations, V is the solution volume, and m is the adsorbent mass. Because both RE and q_e depend on the concentration change $C_0 - C_e$, adsorption capacity can carry strong information about removal efficiency. A model that uses capacity is therefore useful for process monitoring or post-experiment analysis, where capacity has already been measured or estimated. However, the same model should be treated cautiously for pre-experiment screening, because the capacity of a candidate adsorbent under a new condition is usually not known before the experiment is performed. Therefore, capacity-inclusive and capacity-free models answer different practical questions and should not be interpreted in the same way. Recent PFAS adsorption modeling has similarly shown that random-split performance can be much stronger than leave-one-compound-out performance, illustrating that interpolation and transfer to unseen chemical domains are distinct tasks [11].

This study addresses these issues by developing a source-aware and leakage-aware validation framework for AMD-relevant metal-contaminant adsorption removal-efficiency prediction across a broad literature-derived dataset spanning 189 normalized source references, 219 adsorbents, and 16 adsorbate classes. Here, source awareness means tracking the literature reference from which each adsorption record originated, rather than treating all rows as fully independent observations. Leakage awareness means testing whether model performance remains reliable when familiar information is removed from training, and separating feature sets that may be appropriate after an experiment from feature sets that are suitable before an experiment. The framework is intended to support ML-based mathematical modeling for understanding the combined and individual effects of adsorption variables while avoiding unsupported claims of universal deployment.

The framework separates two aspects that are often mixed in adsorption ML studies: what information is available to the model, and how independent the test data are from the training data. Four feature sets were constructed to represent different levels of information availability. A full capacity-inclusive feature set included adsorption capacity and capacity-derived predictors and was interpreted as a process-monitoring or post-experiment analysis model. Three capacity-free feature sets excluded adsorption capacity and were interpreted as pre-experiment screening models, because the capacity of a candidate adsorbent is usually not known before testing. Among the capacity-free models, one represented adsorbates using ion descriptors, one used normalized adsorbate identity as a categorical label, and one used only numeric operating, material, and engineered process variables as a strict screening baseline.

Two validation tracks were then used to separate interpolation from broader generalization. Track A evaluated source-specific interpolation by training and testing within individual literature sources. This asks whether high R^2 values comparable to those reported in the adsorption ML literature can be reproduced inside known source groups. Track B evaluated broader generalization using increasingly strict split designs. In random row-wise validation, individual records were randomly divided between training and testing. In adsorbent-held-out validation, selected adsorbents were kept completely unseen during training and used only for testing. In reference-held-out validation, all records from selected literature sources were kept unseen during training and used only for testing. The novelty of this study is therefore not simply a new regression model, but a source-aware and leakage-aware reporting framework that clarifies when high R^2 reflects valid interpolation within known experimental domains and when additional validation is needed before claiming deployment to new adsorbents, studies, or laboratories.

2. Materials and methods

2.1. Dataset and target variable

The dataset was compiled from literature-reported adsorption experiments for AMD-relevant metal and metalloid contaminants. Each record corresponds to one adsorption condition with a reported removal efficiency. The raw dataset contained 566 rows and 13 columns. After numeric parsing, missing-value filtering, and constraining removal efficiency to the physically meaningful range 0–100%, 563 records remained. The cleaned dataset included 189 normalized source references, 219 adsorbents, and 16 normalized adsorbate classes. The primary modeling domain excluded the sparse neutral/low-concentration regime, leaving 537 records for model training and testing.

The dataset was intended as a heterogeneous multi-source benchmark for evaluating validation design in adsorption ML, rather than as a systematic meta-analysis of all AMD treatment or heavy-metal adsorption studies. Records were retained when removal efficiency and the required operating or material variables were reported or could be deterministically parsed from tabulated values. Records were excluded when essential numeric variables were missing, when removal efficiency could not be constrained to the physical range of 0–100%, or when adsorbent, adsorbate, or source identity could not be normalized consistently.

The target variable was removal efficiency, RE (%). Input variables included adsorbent name, adsorbate identity, surface area, adsorption capacity, initial concentration, contact time, adsorbent dose, agitation speed, initial pH, temperature, and source reference. The source identifier, denoted `Ref_group`, was used as a source-origin proxy because explicit laboratory-origin labels were not available for all records. The pH-concentration regime label was not used as a model target or as a model input feature; it was created only for stratified splitting, diagnostic reporting, and applicability-domain definition.

To interpret validation sensitivity, a source-domain audit was performed on the restricted primary domain. This audit quantified the number of records per normalized source, the number of singleton sources, the fraction of rows contributed by the largest source groups, source-wise RE means and standard deviations, adsorbent and adsorbate diversity within each source, and pH-concentration-regime coverage. A one-way source-label variance diagnostic was also computed as

$$\eta^2_{\text{source}} = \frac{\sum_g n_g (\bar{y}_g - \bar{y})^2}{\sum_i (y_i - \bar{y})^2}, \tag{3}$$

where g indexes source groups, n_g is the number of records in source g, $\bar{y}_g$ is the mean RE for source g, and $\bar{y}$ is the overall mean RE. This statistic was used only to diagnose source imprint in the dataset; `Ref_group` was not used as a predictor in the deployed Track B models.

2.2. Numeric cleaning and categorical normalization

Literature-derived adsorption tables often contain nonuniform numeric formatting. Numeric entries were cleaned using deterministic rules: decimal commas were converted to decimal points, uncertainty and approximate symbols were removed, multiple values in one cell were reduced to the first numeric value, and nonnumeric text was parsed using regular expressions. Rows with missing values in required numeric variables were removed. Adsorbate names were normalized to reduce duplicates such as elemental and valence-state aliases. Adsorbent names were grouped into broad families, including carbon-based, mineral/oxide, polymer, biomass, and composite/other categories.

The cleaned data were partitioned into four pH-concentration regimes using pH = 7 as the acidic/near-neutral boundary and the dataset median initial concentration, $C_0 = 50$ ppm, as the low/high concentration boundary (Table 2). The median C_0 is the middle value of the initial-concentration distribution and provides a data-driven threshold for separating lower- and higher-concentration experiments. These regimes were not used as prediction targets and were not passed to XGBoost, LightGBM, or ExtraTrees as predictor variables. Instead, the models predicted RE (%) from the original experimental and material variables, including numerical pH and numerical initial concentration. Regime labels were used only for stratified random splitting and diagnostic analysis, ensuring that train-test splits preserved chemically meaningful coverage across major operating regions. The acidic regimes dominated the dataset, whereas the neutral/low-C_0 regime contained only 26 observations. Because this regime was too small for stable split-wise evaluation, it was excluded from the primary modeling domain and treated as outside the present applicability domain. The excluded regime was not discarded from the dataset audit; it was reported explicitly as an out-of-domain limitation to avoid presenting unstable validation results from a poorly represented operating region.

Table 2: pH-concentration regime distribution in the cleaned dataset.

Regime	Count	Mean pH	Mean C_0 (ppm)	Mean RE (%)
Acidic_HighC0	246	4.51	190.27	70.30
Acidic_LowC0	231	5.12	11.04	74.06
Neutral_HighC0	60	7.65	166.51	68.05
Neutral_LowC0	26	7.47	20.24	81.41

Neutral_LowC0 was excluded from primary modeling because it contained only 26 observations; the other three regimes formed the 537-record primary domain.

2.3. Feature sets

Four feature sets were constructed to distinguish scientific use cases (Table 3). This separation is important because a model that includes adsorption capacity answers a different question from a model intended for pre-experiment adsorbent screening. Adsorption capacity is closely related to removal efficiency because both are calculated from concentration change during adsorption. In most

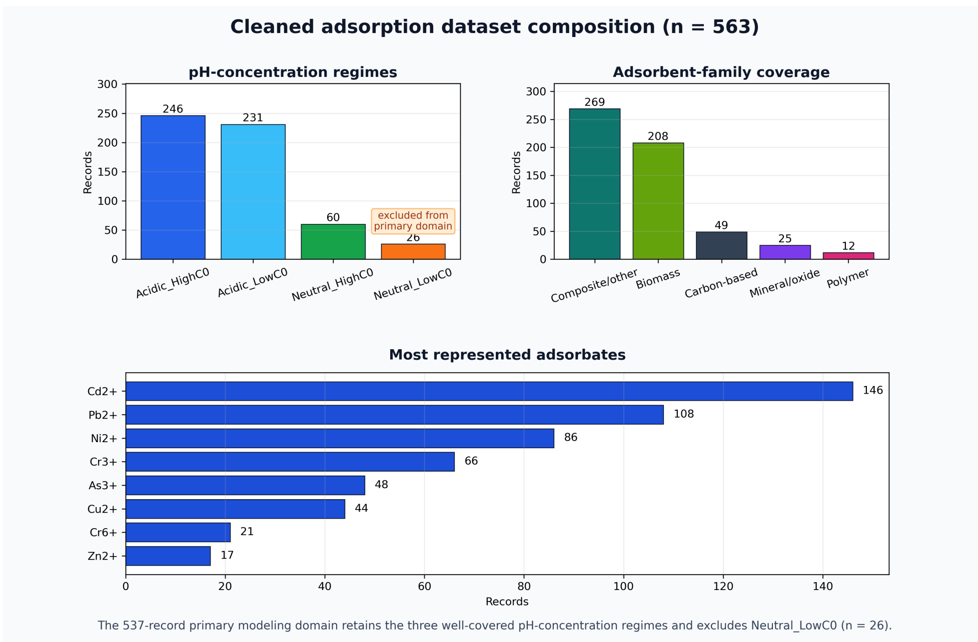

Figure 1: Composition of the cleaned adsorption dataset after deterministic numeric parsing and quality filtering. The sparse neutral/low-C_0 regime was retained in the dataset audit but excluded from the primary 537-record modeling domain.

batch adsorption studies, capacity is not known before an experiment; it is calculated after measuring or estimating equilibrium concentration. Capacity-inclusive models are therefore best interpreted as descriptive, process-monitoring, or post-experiment analysis models. In contrast, capacity-free models use information that is normally available before testing, such as pH, initial concentration, dose, contact time, temperature, surface area, adsorbent family, and adsorbate information. The base experimental inputs were surface area, initial concentration, contact time, dose, agitation speed, initial pH, and temperature. The process-monitoring feature set additionally used adsorption capacity and capacity-derived terms, whereas the screening feature sets deliberately removed these terms.

Physics-inspired variables included site density, logarithmic contact time, mixing index, driving-force proxy, and acidity strength:

$$\text{Site Density} = \frac{\text{dose} \times \text{surface area}}{C_0}, \tag{4}$$

$$\text{LogTime} = \log(1 + \text{contact time}), \tag{5}$$

$$\text{Mixing Index} = \text{RPM} \times \text{LogTime}, \tag{6}$$

$$\text{Driving Force} = \frac{C_0}{\text{dose}}, \tag{7}$$

$$\text{Acidity Strength} = |\text{pH} - 7|. \tag{8}$$

These engineered variables were used as heuristic process descriptors rather than mechanistic adsorption-law parameters. They expose the tree models to simple combinations of dose, surface area, concentration, contact time, agitation, and acidity that are commonly relevant to adsorption behavior.

Ion descriptors were added to represent adsorbates through simple aqueous-chemistry properties rather than only through contaminant-name labels. A one-hot

Table 3: Feature-set definitions used to separate process-monitoring and screening tasks.

Feature set	Included information	Intended use
Full capacity process model	Operating variables, surface area, adsorption capacity, capacity-derived predictors, engineered process variables, ion descriptors, adsorbent family, and adsorbate identity	Descriptive modeling, process monitoring, or post-experiment analysis
Screening-safe plus ion	Operating variables, surface area, engineered process variables, ion descriptors, and adsorbent family; excludes adsorption capacity and capacity-derived terms	Pre-experiment screening with basic ion chemistry
Screening-safe plus adsorbate one-hot	Operating variables, surface area, engineered process variables, normalized adsorbate identity, and adsorbent family; excludes adsorption capacity and capacity-derived terms	Pre-experiment screening when contaminant identity is known
Screening-safe numeric-only	Operating variables, surface area, and engineered process variables; excludes adsorption capacity, capacity-derived terms, and direct adsorbate identity	Strict screening baseline

adsorbate label identifies the metal species but does not encode chemical similarity between ions. Descriptors such as charge, ionic radius, hydrated radius, ionic potential, and hydration ratio allow the model to partially capture differences in ion size, charge density, and hydration behavior that can influence adsorption.

The full model therefore used the following ML inputs: seven operating/material numeric variables; adsorption capacity, capacity-loading ratio, and thermo-capacity; five engineered process descriptors; six ion descriptors; normalized adsorbate identity; and broad adsorbent family. The capacity-free models retained the operating/material variables and engineered descriptors but removed adsorption capacity, capacity-loading ratio, and thermo-capacity. Among the capacity-free models, one retained ion descriptors, one used normalized adsorbate identity, and the strictest model used numeric variables only.

Capacity-derived predictors were included only in the capacity-inclusive process model and were excluded from all screening-safe models. They were defined as

$$\text{Capacity Loading Ratio} = \frac{q_e}{C_0}, \tag{9}$$

$$\text{Thermo Capacity} = q_e T, \tag{10}$$

where q_e is adsorption capacity, C_0 is initial concentration, and T is temperature.

2.4. Machine-learning models

XGBoost was selected as the primary model because gradient-boosted trees are effective for nonlinear tabular datasets and are widely used in adsorption ML studies [12, 13]. LightGBM was included as an additional gradient-boosting comparator because preliminary model development indicated similar row-wise interpolation performance [14]. Extremely randomized trees (ExtraTrees) were included as a comparator ensemble model because random forest-style tree ensembles provide robust nonlinear baselines for tabular prediction [15, 16]. A mean dummy regressor was included to quantify improvement beyond a no-skill baseline.

Categorical variables were one-hot encoded with unknown-category handling, while numerical variables were passed directly to the tree models. For each split, categorical encoding and model fitting were performed within the training fold only, and the fitted preprocessing pipeline was then applied to the corresponding test fold. Unknown categories appearing only in test splits were ignored by the encoder rather than fitted using test information. This ensured that preprocessing did not introduce information from the test set into the training process.

XGBoost used 140 estimators, learning rate 0.05, maximum depth 4, subsample 0.85, column subsample 0.85, ℓ_1 regularization 0.1, and ℓ_2 regularization 1.0. The standard LightGBM model used 300 estimators, learning rate 0.05, maximum depth 5, 31 leaves, minimum child samples 15, subsample 0.85, column subsample 0.85, ℓ_1 regularization 0.1, and ℓ_2 regularization 0.5. ExtraTrees used 120 estimators, maximum depth 10, and minimum leaf size 2.

Two additional LightGBM sensitivity configurations were evaluated because preliminary model development had produced strong single-split LightGBM performance. These configurations were not used to claim a newly optimized algorithm; they were included as sensitivity checks to test whether the validation conclusions were stable across reasonable gradient-boosting complexity settings. The conservative configuration used 600 estimators, learning rate 0.03, maximum depth 5, 20 leaves, minimum child samples 25, subsample 0.80, column subsample 0.80, ℓ_1 regularization 0.5, and ℓ_2 regularization 1.0. The expanded configuration used 500 estimators, learning rate 0.035, maximum depth 6, 40 leaves, minimum child samples 10, subsample 0.90, column subsample 0.90, ℓ_1 regularization 0.05, and ℓ_2 regularization 0.2. Predictions were clipped to the physically meaningful RE range of 0–100% before metric calculation. This clipping was applied only to final model predictions and did not affect training targets.

2.5. Validation design

Two validation tracks were used (Fig. 2). Track A evaluated source-specific interpolation. Although the cleaned dataset contained 189 normalized source references, most references contributed only a small number of rows. Therefore, source-specific 5-fold cross-validation was reported only for references with at least 20 usable records and sufficient variation in removal efficiency, using shuffled folds with a fixed random seed. Here, "5-fold" refers to the number of cross-validation folds within each eligible source group, not to the number of source groups reported. This tests whether high R^2 values are achievable within a known experimental source, which is similar to the validation setting of many high-performance adsorption ML studies.

Track B evaluated global generalization using three validation protocols within the restricted primary domain. The random row-wise split used an 80/20 train-test split stratified by the three retained pH-concentration regimes, so each split retained acidic/high-C_0, acidic/low-C_0, and neutral/high-C_0 observations. This stratification affected only how rows were assigned to training and test sets; the regime label itself was not included in the fitted model. The adsorbent-held-out split used group splitting so that exact adsorbent material names in the test set were absent from the training set; all rows for a held-out adsorbent identity were assigned to the test split. This protocol tests transfer to unseen material identities when measured descriptors such as surface area, operating conditions, pH, dose, and concentration are available. It does not claim prediction for a completely undescribed material from name alone. The reference-held-out split used group splitting so that entire references/sources were absent from the training set, making it the stricter proxy for transfer across literature sources, experimental protocols, adsorbent synthesis routes, concentration windows, contaminant distributions, and reporting practices. Because all data were literature-derived, reference-held-out validation was used as a proxy for external-source validation and should not be interpreted as a substitute for future prospective laboratory validation. Each Track B protocol was repeated over five random seeds. Metrics included R^2, root mean squared error (RMSE), mean absolute error (MAE), and prediction bias.

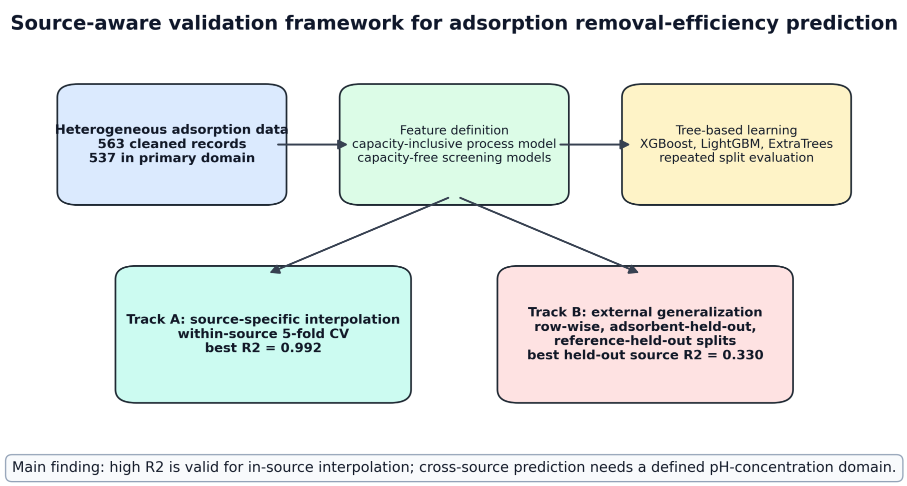

Figure 2: Workflow of the proposed source-aware adsorption-ML validation framework. Track A measures source-specific interpolation, whereas Track B measures progressively stricter global generalization.

3. Results

The results are reported in the same order as the validation framework: first, source-specific interpolation within coherent literature sources; second, repeated random row-wise prediction in the restricted primary domain; third, stricter adsorbent-held-out and reference-held-out transfer; and fourth, comparison of capacity-inclusive and capacity-free feature sets. This ordering separates interpolation accuracy from broader external-source generalization.

3.1. Source-specific models reproduced high adsorption-prediction accuracy

Track A showed that high removal-efficiency prediction accuracy is achievable when models are trained and evaluated within individual source groups. The source groups in Table 4 are not the only sources in the dataset; they are the clean bibliographic source groups that met the minimum row-count and target-variability criteria and produced stable positive within-source cross-validation performance. The full capacity process model achieved R^2 = 0.9916 for the source group Shen et al. (2017b), R^2 = 0.9735 for Gao et al. (2019), and R^2 = 0.9601 for Zama et al. (2017). These results are in the range commonly reported in adsorption ML studies under row-wise or within-domain validation [3, 5, 4]. They are also chemically plausible because the dominant source groups are internally coherent adsorption campaigns rather than arbitrary row collections: Shen et al. studied Ni(II) adsorption on wheat-straw-pellet and rice-husk biochars, Gao et al. investigated Cd(II) adsorption mechanisms on rice-straw and sewage-sludge biochars, Zama et al. studied Pb(II), Cd(II), and As(III) sorption/desorption across biochars, and Cui et al. focused on Cd(II) removal by *Canna indica*-derived biochar [17, 18, 19, 20]. Two additional ScienceDirect-indexed URL-bucket groups met the row-count threshold but were not treated as clean source-specific Track A groups because their reference metadata indicated mixed or unresolved bibliographic grouping. They are retained in the benchmark audit trail, but not used to support the high-interpolation claim. This reinforces the central point that source-specific predictability should be evaluated only for coherent source groups and should not be summarized by a single global accuracy value.

Table 4: Track A source-specific 5-fold cross-validation results for the four coherent source groups with stable positive within-source predictability. Complete Track A benchmark outputs, including unresolved ScienceDirect URL-bucket groups, are retained in the submitted CSV files for auditability.

Source group	Rows	R^2	RMSE	MAE
Shen et al. (2017b)	65	0.9916	3.20	2.33
Gao et al. (2019)	48	0.9735	2.97	2.14
Zama et al. (2017)	91	0.9601	7.84	3.56
Cui et al. (2016a)	24	0.8936	6.65	4.56

Capacity-free source-specific models also performed well in selected source groups. For example, the source-specific model for Zama et al. (2017) achieved $R^2 = 0.9250$ using the screening-safe ion feature set, while Shen et al. (2017b) achieved R^2 values above 0.91 using capacity-free screening feature sets. This indicates that high source-specific performance is not solely a capacity-inclusion artifact, although capacity-inclusive models generally gave the strongest interpolation performance.

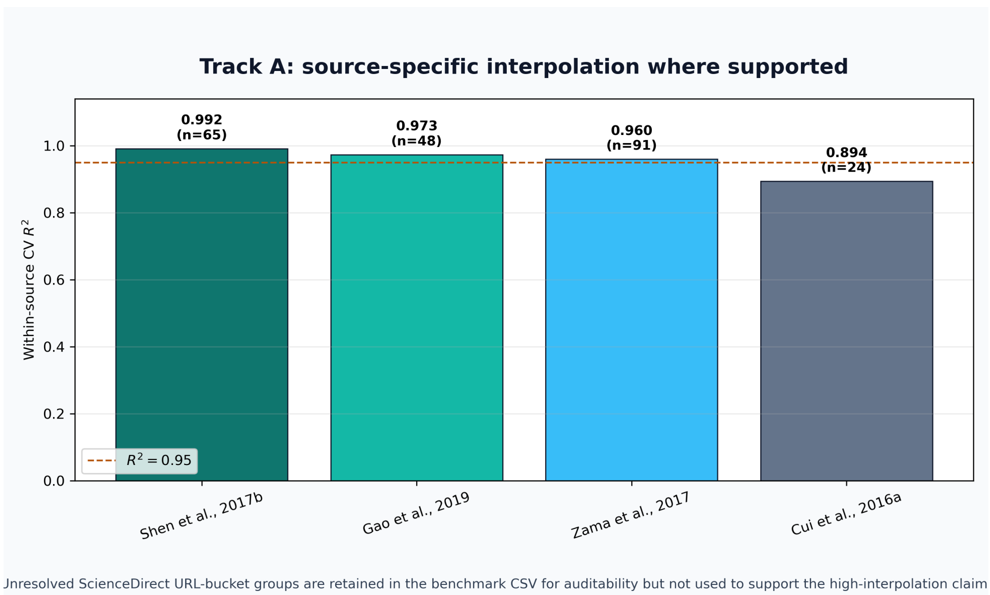

Figure 3: Track A source-specific interpolation performance. Several individual source-group models achieved R^2 values near or above 0.95, while other source groups remained more difficult.

3.2. Random row-wise validation produced moderate-to-good global performance

Under repeated random row-wise splits in the restricted primary domain, the full capacity process model achieved R^2 = 0.8064 and RMSE = 12.94 with LightGBM, while the corresponding XGBoost model achieved R^2 = 0.7985 and RMSE = 13.22. The screening-safe plus adsorbate one-hot XGBoost model achieved R^2 = 0.7283 and RMSE = 15.29, while the screening-safe plus ion XGBoost model achieved R^2 = 0.7265 and RMSE = 15.34. Thus, even without adsorption capacity, the model retained useful predictive ability under in-dataset interpolation (Table 5).

Table 5: Restricted-domain random row/regime split performance after excluding the sparse neutral/low-C_0 regime. Values are means over five repeated splits.

Feature/model	R^2	RMSE	MAE
Full capacity LightGBM	0.8064	12.94	8.08
Full capacity XGBoost	0.7985	13.22	8.44
Full capacity ExtraTrees	0.7730	14.02	9.02
Screening-safe plus adsorbate XGBoost	0.7283	15.29	10.84
Screening-safe plus ion XGBoost	0.7265	15.34	10.88

These results support the use of ML for in-domain adsorption prediction. They also explain the strong preliminary LightGBM results obtained during model development. Earlier LightGBM reports used a single random/regime-stratified split and included capacity-related predictors, giving test R^2 = 0.847 and RMSE = 12.05; in the present repeated-split benchmark, individual LightGBM random splits similarly reached R^2 = 0.8499 for the standard configuration and

$R^2 = 0.8570$ for the conservative sensitivity configuration. However, the five-split means were lower ($R^2 = 0.8064$ and $R^2 = 0.7983$, respectively), showing that the earlier value was a valid favorable row-wise interpolation result rather than a stable estimate of source-held-out generalization. Random row-wise validation alone is therefore not sufficient evidence for deployment on unseen sources.

The two capacity-free XGBoost screening models differed in how the adsorbate was represented. The screening-safe plus adsorbate model used normalized adsorbate identity as a categorical one-hot feature, allowing the model to learn contaminant-specific effects directly. The screening-safe plus ion model used simple aqueous-ion descriptors, including charge, ionic radius, hydrated radius, ionic potential, and hydration ratio, instead of the direct contaminant label. The similar row-wise performances of these two models, $R^2 = 0.7283$ and $R^2 = 0.7265$, respectively, indicate that basic ion-chemistry descriptors captured much of the information provided by explicit adsorbate labels in the random-split setting.

3.3. Cross-material and cross-source validation tested broader transfer

The validation protocol strongly affected performance, which is expected for a heterogeneous dataset spanning many source references, adsorbent identities, and metal ions. In the adsorbent-held-out protocol, all experiments associated with selected adsorbent material names were removed from the training set and used for testing. This is different from holding out adsorbates: the held-out groups are adsorbent materials, not contaminant ions. Under this cross-material test, the best capacity-free result was obtained by the screening-safe numeric expanded-LightGBM model, with $R^2 = 0.5323$ and RMSE = 19.35. The screening-safe numeric XGBoost model achieved $R^2 = 0.5199$, while the conservative full-capacity LightGBM model achieved $R^2 = 0.5258$. These results indicate moderate but imperfect transfer to unseen adsorbent identities when measured descriptors and operating conditions are available.

Reference-held-out validation was the most important broad-source stress test because every row from selected literature references was kept outside training. The model was therefore evaluated on source domains whose experimental protocol, adsorbent preparation, concentration window, contaminant distribution, and reporting pattern were not represented by the same reference during training. This validation is closer to the practical question of whether a literature-trained model transfers to a new study. It was more difficult than row-wise validation but improved markedly after removing the sparse neutral/low-C_0 regime. The best reference-held-out sensitivity result was obtained by the full-capacity expanded-LightGBM model, with $R^2 = 0.3296$ and RMSE = 22.88. Because this configuration includes capacity-related predictors, it should be interpreted as a process-monitoring or post-experiment result rather than as a pre-experiment screening result. The best capacity-free reference-held-out result remained the screening-safe numeric-only ExtraTrees model, with $R^2 = 0.3054$ and RMSE = 23.54. These results remain below row-wise performance, but they show that defining a better-covered applicability domain produces more meaningful external-source validation for a broad, multi-source adsorption dataset.

These results show that removal-efficiency prediction is strongly source-dependent. The same data and model family can produce high R^2 under source-specific interpolation, moderate-to-good R^2 under random global validation, and lower but still informative R^2 under source-held-out validation when the applicability domain is restricted to sufficiently represented pH-concentration regimes.

Table 6: Restricted-domain Track B generalization performance after excluding the sparse neutral/low-C_0 regime. Values are means over five repeated splits.

Validation	Feature/model	R^2	RMSE	MAE
Holdout adsorbent	Screening-safe numeric expanded LightGBM	0.5323	19.35	14.55
Holdout adsorbent	Full capacity conservative LightGBM	0.5258	19.19	13.00
Holdout adsorbent	Screening-safe numeric-only XGBoost	0.5199	19.61	15.15
Holdout reference/source	Full capacity expanded LightGBM	0.3296	22.88	17.67
Holdout reference/source	Screening-safe numeric-only ExtraTrees	0.3054	23.54	18.67
Holdout reference/source	Full capacity standard LightGBM	0.2820	23.78	18.67

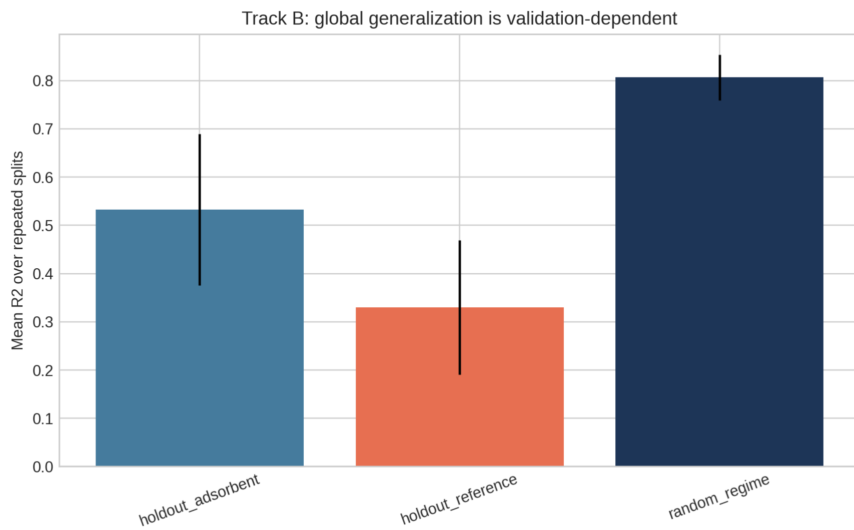

Figure 4: Track B validation comparison. Row-wise validation, adsorbent-held-out validation, and reference-held-out validation represent increasingly strict generalization tasks.

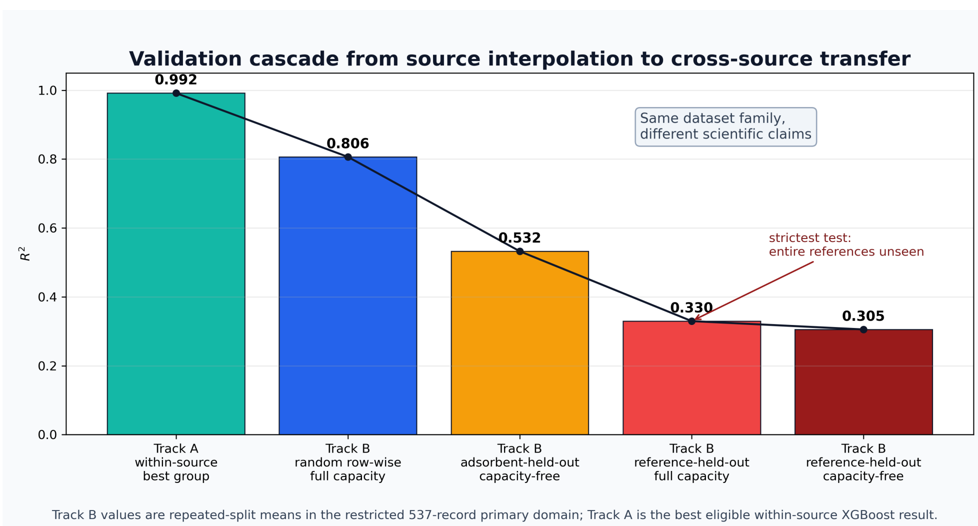

Figure 5: Validation cascade from source-specific interpolation to stricter global transfer. The figure summarizes why high R^2 values near 0.95 are achievable within known sources while reference-held-out prediction is a more difficult external-source task.

3.4. Source-domain heterogeneity explained validation sensitivity

The decrease from source-specific or row-wise performance to reference-held-out performance was not simply an algorithmic failure; it reflected the structure of the literature-compiled dataset. In the 537-record primary domain, the records were distributed across 180 normalized source references, but 142 of those sources contributed only one record and the median source size was one row (Table 7). Only six sources contributed at least 20 records, yet those six sources accounted for 52.7% of the primary-domain rows. Therefore, a random row-wise split can place closely related experiments from the same source campaign in both training and testing, whereas reference-held-out validation deliberately removes that source-specific support.

This audit identifies the hidden assumption behind the performance drop. Row-wise validation implicitly assumes that individual rows are exchangeable samples from one common adsorption distribution. The compiled dataset violates that assumption: source groups differ in contaminant identity, adsorbent synthesis route, pH and concentration window, measured material descriptors, analytical protocol, and reporting convention. The source-label variance diagnostic attributed 42.4% of the RE variance to source grouping, and the spread of source-wise mean RE values was larger than the median within-source spread. Thus, reference-held-out validation is not merely a harsher score; it tests whether a model can transfer when source-specific experimental context is no longer shared between training and testing.

3.5. Capacity-inclusive and capacity-free models answered different questions

The full capacity process model performed best for random row-wise validation and several source-specific models. This is expected because adsorption capacity

Table 7: Source-domain heterogeneity diagnostics for the 537-record primary modeling domain.

Diagnostic	Value
Primary-domain records	537
Normalized source references	180
Singleton source references	142
Median records per source reference	1
Sources with at least 20 records	6
Rows contributed by six largest source groups	52.7%
Overall RE mean ± standard deviation	71.66 ± 29.92 percentage points
Standard deviation of source-wise RE means	19.26 percentage points
Median within-source RE standard deviation	7.37 percentage points
Source-label variance diagnostic, η^2_{source}	0.424

The source-label variance diagnostic is descriptive and was not used as a model input. Because many references were singletons, η^2_{source} should be interpreted as evidence of source imprint rather than as a causal partition of adsorption mechanisms.

carries strong information about adsorption performance. However, adsorption capacity is derived after measuring concentration change and is usually not available before testing a candidate adsorbent under a new condition. Therefore, capacity-inclusive results should be interpreted as descriptive, process-monitoring, or post-experiment analysis results. Capacity-free models are more appropriate for pre-experiment screening and adsorbent-design guidance because they exclude adsorption capacity and capacity-derived predictors. In the restricted-domain adsorbent-held-out and reference-held-out tests, capacity-free numeric models were competitive or superior in several cases. This should not be interpreted as adsorption capacity being uninformative; rather, it suggests that capacity-related predictors can improve interpolation while also encoding source- or condition-specific relationships that do not always transfer cleanly across held-out materials or references.

4. Discussion

4.1. Why near-0.95 R^2 is achievable but validation-dependent

The Track A results explain why many adsorption ML studies report R^2 values near or above 0.95. Within a single source, adsorbent family, contaminant system, or experimental protocol, the response surface can be learned accurately by tree-based models. In this setting, high R^2 is scientifically plausible and not automatically invalid. The present results reproduce that behavior, with source-specific R^2 values of 0.960–0.992 for three major source groups.

However, Track B shows that this performance should not be interpreted as universal adsorption prediction. When sources are held out, the model must transfer across experimental protocols, adsorbent synthesis methods, contaminant distributions, concentration ranges, and reporting conventions. Under that validation setting, performance fell below row-wise performance even after excluding the sparse neutral/low-C_0 regime. The result is consistent with recent adsorption-related work showing that random-split performance can be substantially higher than leave-compound-out performance [11].

The lower reference-held-out R^2 should therefore be read as a scientific signal rather than only a modeling weakness. It indicates that important context is missing from the current descriptor set. Likely missing variables include pH at point of zero charge, pore volume, pore-size distribution, surface functional groups, zeta potential, cation-exchange capacity, feedstock and pyrolysis details for biochars, aqueous speciation, ionic strength, competing ions, and laboratory protocol metadata. Without such descriptors, the model can learn source-specific patterns under interpolation but has limited information for explaining why a new source behaves differently.

4.2. Feature availability changes the scientific claim

The comparison between capacity-inclusive and capacity-free feature sets shows that feature availability changes the meaning of the prediction task. Adsorption capacity carries strong information about removal efficiency because both are derived from concentration change during adsorption. Capacity-inclusive models are therefore useful for descriptive modeling, process monitoring, and post-experiment analysis, but they should not be interpreted as purely prospective adsorbent-screening models unless capacity is already known or estimated independently.

In contrast, capacity-free models exclude adsorption capacity and capacity-derived terms, making them more appropriate for pre-experiment screening. These models rely on information that is typically available before testing, such as pH, initial concentration, contact time, dose, temperature, surface area, adsorbent family, and adsorbate information. The similar row-wise performance of the adsorbate-label and ion-descriptor screening models indicates that simple ion-chemistry descriptors can capture much of the information provided by explicit contaminant labels in the random-split setting. The competitive performance of capacity-free numeric models under held-out validation further suggests that simpler descriptor sets may sometimes transfer more robustly across materials or sources, even if capacity-inclusive models remain stronger for within-domain interpolation.

4.3. Relation to existing adsorption ML literature

The proposed framework complements rather than contradicts previous high-accuracy adsorption ML studies. Hafsa et al. demonstrated that tree-based algorithms can achieve high removal-efficiency prediction accuracy for heavy-metal adsorption datasets under repeated cross-validation [3]. Wang et al. showed that XGBoost, SHAP, and partial-dependence analyses can identify interpretable drivers of heavy-metal removal by biochar [5]. Lu et al. showed that random forests can predict heavy-metal removal by chitosan-based flocculants with R^2 above 0.93 [4]. Barkhordari et al. further demonstrated that interpretable XGBoost models are relevant to JECE's environmental-remediation scope by predicting heavy-metal removal efficiency in electrokinetic remediation [7]. These studies establish that ML is useful for environmental treatment modeling.

The present study adds a validation-layer contribution: adsorption ML workflows should explicitly state whether they are predicting within a known source or generalizing to unseen sources. This distinction is critical because high row-wise R^2 can be overoptimistic when the deployment target is a new laboratory, adsorbent synthesis route, contaminant matrix, or literature source. The broader ML literature has identified leakage and validation mismatch as major reproducibility risks [9, 10], and adsorption datasets have several mechanisms by which such mismatch can occur.

4.4. Practical implications for adsorption model deployment

For practical adsorption screening, a single universal RE model should be used cautiously. A more defensible workflow is to define the pH-concentration applicability domain before modeling, exclude or separately model poorly represented regimes, use source-specific or laboratory-calibrated models for high-accuracy optimization within a known experimental system, use capacity-free screening models for preliminary adsorbent and operating-condition selection, and report source-held-out validation when predictions are intended for new adsorbents, contaminants, laboratories, or literature domains. This framing allows high-R^2 models to remain useful without overstating their generality.

The most direct route to improving cross-source performance is not simply replacing XGBoost or LightGBM with a more complex learner. The source audit suggests that external performance would more likely improve through richer mechanistic descriptors, better source metadata, and validation-aware modeling. Future versions of the dataset should prioritize adsorbent surface chemistry, synthesis and activation conditions, solution chemistry, and explicit laboratory/source labels. Once those descriptors are available, hierarchical or multi-task models could separate universal adsorption trends from source-specific offsets, while still reporting reference-held-out or laboratory-held-out validation as the primary deployment test.

4.5. Limitations and future work

This study has several limitations. First, the dataset represents adsorption tests for AMD-relevant metal and metalloid contaminants, but many records are single- or limited-ion aqueous experiments rather than complete real AMD matrices. Therefore, sulfate concentration, ionic strength, competing ions, dissolved organic matter, and site-specific mine-water composition should be added before claiming direct deployment to complex field AMD. Second, the dataset does not contain explicit laboratory-origin labels for all records; bibliographic reference was therefore used as a source proxy. Future work should annotate records as in-house, external-laboratory, or literature-derived and repeat validation with leave-laboratory-out splits. Third, the ion descriptor table should be expanded with fully cited descriptors such as electronegativity, hydration energy, softness/hardness, and likely aqueous speciation. Fourth, important adsorbent descriptors such as pH_{pzc}, pore volume, pore size, zeta potential, FTIR-derived functional groups, and cation-exchange capacity are incomplete in the present dataset. Fifth, the neutral/low-concentration regime contained only 26 observations and was excluded from primary modeling; predictions in that regime should therefore be treated as outside the present applicability domain until more data are available. Sixth, because all records were literature-derived, reference-held-out validation should be interpreted as a proxy for external-source transfer, not as a substitute for prospective laboratory validation. Finally, the dataset was designed as a heterogeneous ML benchmark for source-aware validation rather than as a complete systematic meta-analysis of all AMD treatment or heavy-metal adsorption studies. These limitations define clear routes for improving external predictive performance while preserving the source-aware validation principle.

5. Conclusions

This study developed a source-aware and leakage-aware framework for AMD-relevant metal-contaminant adsorption removal-efficiency prediction. High R^2 values were achievable within individual adsorption sources, with source-specific XGBoost models reaching R^2 up to 0.992. Because the neutral/low-concentration regime contained only 26 observations, primary global modeling excluded that sparse regime and used a 537-record restricted domain. Within this domain, random row-wise validation reached $R^2 = 0.806$ for the full capacity LightGBM process model, adsorbent-held-out validation reached $R^2 = 0.532$ for the capacity-free numeric expanded-LightGBM screening model, and reference-held-out validation reached $R^2 = 0.330$ for the full-capacity expanded-LightGBM sensitivity model. The best capacity-free reference-held-out model reached $R^2 = 0.305$. These findings show that high-accuracy interpolation, applicability-domain definition, and cross-source generalization are different scientific tasks. High adsorption-ML accuracy is meaningful when the claim is interpolation within known sources or well-represented operating domains, but it is not sufficient by itself to claim deployment to unseen adsorbents, unseen studies, or new laboratories. Future adsorption ML studies should report all three, separate capacity-inclusive process-monitoring models from capacity-free screening models, and include source- or laboratory-held-out validation when deployment beyond the training source is claimed.

CRediT authorship contribution statement

Yash Bisht: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft. Margreth Tadie: Supervision, Validation, Writing – review and editing. Susmit Chitransh: Data curation, Conceptualization, Investigation, Validation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The curated dataset, analysis scripts, and generated benchmark outputs are available from the corresponding author upon reasonable request and will be deposited in a public repository before final publication.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI's Codex to assist with code organization, LaTeX formatting, and language polishing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the manuscript.

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